THE COLOURED SAND THEOREM: THE p-PART OF THE SANDPILE GROUP OF THE HAMMING GRAPH

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ABSTRACT. Let $H(d,q) = K_q^d$ be the Hamming graph and $K(H(d,q))$ its sandpile group (critical group). For primes $p \nmid q$ its p-part was determined by Jacobson, Niedermaier and Reiner in 2003, who called the remaining primes difficult, "due to bad behavior of its p-primary structure". We treat the primes $p \mid q$. Jacobson, Niedermaier and Reiner showed that over $\mathbb{Z}$ the Laplacian splits into explicit triangular pieces; we read these pieces as operators $q(c+H)+U$ on the Boolean lattice (A). From this we prove: (B) a closed formula for the Sylow p-subgroup for every prime $p > d$ and every q divisible by p, in which ballot numbers appear; (C) the exponent of the Sylow p-subgroup for every prime p and every $q \geq 3$, namely $p^{\max_{1\leq s\leq d}(es+v_p(s))}$ with $e = v_p(q) \geq 1$; (D) the number of its cyclic factors when $p \mid q$, from Wilson's diagonal form of inclusion matrices. For $p \leq d \leq 2p-2$ we state a conjecture, the Glue Conjecture, which describes the group exactly; it was found from the cases $p = 3, 5$ and then predicted, before they were computed, 44 further cases, all correctly.

1. INTRODUCTION

For a connected graph Γ with Laplacian $L(\Gamma)$, the sandpile group $K(\Gamma)$ is the torsion subgroup of $\operatorname{coker} L(\Gamma)$, and $\operatorname{coker} L(\Gamma) \cong \mathbb{Z} \oplus K(\Gamma)$; its order is the number of spanning trees [Big99]. The Hamming graph $H(d,q)$ has vertex set $(\mathbb{Z}/q)^d$, two vertices being adjacent when they differ in exactly one coordinate; it is the Cartesian power K_q^d of the complete graph. Its Laplacian eigenvalues are qs, $0 \leq s \leq d$, with multiplicity $\binom{d}{s}(q-1)^s$.

Jacobson, Niedermaier and Reiner [JNR03] determined the Sylow p-subgroup of $K(K_{n_1} \times \cdots \times K_{n_k})$ for every prime p that divides none of the n_i [JNR03, Theorem 2], and bounded the number of invariant factors [JNR03, Theorem 3]. About the other primes they wrote: "Describing the exact structure of the critical group $K(K_{n_1} \times \cdots \times K_{n_k})$ appears to be difficult, due to bad behavior of its p-primary structure for primes p that divide some of $n_1, \ldots, n_k$." Ducey and Jalil [DJ14] recovered the results of [JNR03] by another method. For $q = 2$ the graph is the hypercube; its odd part is due to Bai [Bai03] and its 2-part was determined in [DST].

In this paper $q \geq 2$ is arbitrary, p is a prime dividing q, and $e := v_p(q) \geq 1$, except where we say otherwise.

The pieces. Let B_m be the free abelian group on the subsets of $[m] = \{1, \ldots, m\}$, graded by size: $B_m = \bigoplus_{s=0}^{m} B_m(s)$. Let H act on $B_m(s)$ as the scalar s, and let U and D be the up and down operators,

$$U(S) = \sum_{x \notin S} S \cup \{x\}, \qquad D(S) = \sum_{x \in S} S \setminus \{x\}.$$

For integers $c \geq 0$ put

$$X_q(m,c) := \operatorname{coker}\big(q(c+H)+U \ \text{ on } B_m\big).$$

It is finite when $c \geq 1$; for $c = 0$ it is $\mathbb{Z}$ plus a finite group.

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Theorem A (Jacobson–Niedermaier–Reiner [JNR03, Theorem 13])**.** *For every $q \geq 2$ and $d \geq 1$, as abelian groups,*

$$\operatorname{coker} L\big(H(d,q)\big) \;\cong\; \bigoplus_{j=0}^{d} X_q(d-j,\, j)^{\oplus \binom{d}{j}(q-2)^j}.$$

This is the case $n_1 = \cdots = n_d = q$ of [JNR03, Theorem 13]: there the pieces are upper triangular matrices $M(N; n_S)$ of size $2^{|S|}$ with identity blocks above the diagonal, and after reversing the order of the subsets of S they are the matrices $q(c + H) + U$. We include the short proof below, in the form we use. For $q = 2$ only $j = 0$ survives and the theorem says $\operatorname{coker} L(Q_d) \cong X_2(d, 0)$, the starting point of [DST].

Theorem B (the primes $p > d$)**.** *Let $d \geq 1$, let $p > d$ be a prime dividing q, and $e = v_p(q)$. Then*

$$\operatorname{Syl}_p K\big(H(d,q)\big) \;\cong\; \bigoplus_{v=1}^{d} \big(\mathbb{Z}/p^{ev}\big)^{n_v}, \qquad n_v = \sum_{\substack{j+2k=d+1-v \\ (j,k)\neq(0,0)}} \binom{d}{j} (q-2)^j\, b(k,v),$$

the sum over integers $j, k \geq 0$, where $b(k,v) = \frac{v}{v+2k}\binom{v+2k}{k} = [x^k]\, C(x)^v$ are the ballot numbers and $C(x) = \sum_n \frac{1}{n+1}\binom{2n}{n} x^n$ is the Catalan series.

Equivalently, with $c(z) = C(z^2)$ and $u = q - 2$, $\sum_v n_v y^v = [z^{d+1}]\,(1+uz)^d\, \frac{yz\,c(z)}{1-yz\,c(z)} - y^{d+1}$. For instance, with $u = q - 2$:

$$\begin{aligned}
d = 2: &\quad (\mathbb{Z}/p^e)^{u^2+1} \oplus (\mathbb{Z}/p^{2e})^{2u},\\
d = 3: &\quad (\mathbb{Z}/p^e)^{u^3+3u} \oplus (\mathbb{Z}/p^{2e})^{3u^2+2} \oplus (\mathbb{Z}/p^{3e})^{3u},\\
d = 4: &\quad (\mathbb{Z}/p^e)^{u^4+6u^2+2} \oplus (\mathbb{Z}/p^{2e})^{4u^3+8u} \oplus (\mathbb{Z}/p^{3e})^{6u^2+3} \oplus (\mathbb{Z}/p^{4e})^{4u}.
\end{aligned}$$

Theorem B is the case $m < p$ of a statement about single pieces (Theorem E in Section 3), which also gives, for every prime p, the contribution of all pieces with $d - j < p$.

Theorem C (the exponent)**.** *Let $q \geq 3$, let p be any prime and $e = v_p(q)$. The exponent of $\operatorname{Syl}_p K(H(d,q))$ is p^E, where*

$$E = \max_{1\leq s\leq d} \big(e\,s + v_p(s)\big) \quad \text{if } e \geq 1, \qquad E = \lfloor \log_p d \rfloor \quad \text{if } e = 0.$$

For $p > d$ this gives $E = ed$, as in Theorem B. For $p \leq d$ the maximum need not be at $s = d$: for $q = 3$ and $d = 10$ it is at $s = 9$ and $E = 11$.

Theorem D (the number of cyclic factors)**.** *Let $p \mid q$. The number of cyclic factors of $\operatorname{Syl}_p K(H(d,q))$ is*

$$N = \sum_{j=0}^{d} \binom{d}{j} (q-2)^j \Big(2^{\,d-j} - \rho_p(d-j)\Big) \;-\; 1, \qquad \rho_p(m) = \sum_{s=0}^{m-1} r_p(m,s),$$

where $r_p(m,s) = \sum_i \big(\binom{m}{i} - \binom{m}{i-1}\big)$, the sum over $0 \leq i \leq t$ with $p \nmid k - i$, and $(t,k) = (s, s+1)$ if $2s \leq m-1$, $(t,k) = (m-1-s, m-s)$ otherwise.

Here $\rho_p(m)$ is the rank over $\mathbb{F}_p$ of U on B_m. For $q = p = 3$ the theorem gives $N = 1, 4, 12, 34, 96, 274, \ldots$, and one checks from the formula that $N = 3^{d-1} + 2^{d-1} - 1$ for $d \leq 40$ (we have not proved this closed form in general). For $q = p$ the value of N equals the bound $\frac{1}{2}(q^d + (q-2)^d) - 1$ of [JNR03, Theorem 3] for $d \leq 2$, is one less for $d = 3$, and is strictly smaller for $4 \leq d \leq 8$ and $p \leq 11$ (where we compared).

Corollary 1. *$\operatorname{Syl}_p K(H(d,q))$ depends on q only through $e = v_p(q)$ and the multiplicities $\binom{d}{j}(q-2)^j$.*

Status. This is a preprint. The proofs of Theorems B–D and E were re-read once by the system that wrote them (see Provenance) and checked against the computations of Section 7; they have not yet been read by an independent reader. Conjecture F is supported by computation only.

What remains is the range $d \geq p$, and there only the pieces $X_q(m,c)$ with $m \geq p$ (Theorem E). For $m \leq 2p-2$ we state a precise conjecture in Section 6, with its evidence; beyond that range we leave the question open.

2. The splitting (after Jacobson, Niedermaier and Reiner)

Proof of Theorem A. Let $K_q = qI - J$ be the Laplacian of the complete graph K_q, acting on $\mathbb{Z}^q$ with basis $e_0, \dots, e_{q-1}$, and put $f_i = e_i - e_0$ $(1 \leq i \leq q-1)$ and $\sigma = f_1 + \cdots + f_{q-1}$. Then $e_0, \sigma, f_1, \dots, f_{q-2}$ is a basis of $\mathbb{Z}^q$ (unimodular change from $e_0, f_1, \dots, f_{q-1}$), and

$$K_q e_0 = -\sigma, \qquad K_q \sigma = q\sigma, \qquad K_q f_i = q f_i.$$

So $\mathbb{Z}^q = A \oplus W$ with $A = \mathbb{Z}e_0 \oplus \mathbb{Z}\sigma$ and $W = \bigoplus_{i \leq q-2} \mathbb{Z} f_i$, both stable under K_q, and K_q is the scalar q on W. The Laplacian of $H(d,q)$ is $L = \sum_{i=1}^{d} 1 \otimes \cdots \otimes K_q \otimes \cdots \otimes 1$ on $(\mathbb{Z}^q)^{\otimes d} = \bigoplus_{T \subseteq [d]} A^{\otimes([d]\setminus T)} \otimes W^{\otimes T}$, and each summand is L-stable. On the summand of T with $|T| = j$, L acts as $(L_A + qj) \otimes 1$, where $L_A = \sum_i K_q^{(i)}$ on $A^{\otimes(d-j)}$ and $W^{\otimes T} \cong \mathbb{Z}^{(q-2)^j}$. There are $\binom{d}{j}$ sets T of size j.

Index the basis of $A^{\otimes m}$ by the subsets $S \subseteq [m]$, the factor in position i being σ if $i \in S$ and e_0 otherwise. Then $L_A(S) = q|S|\, S - \sum_{x \notin S} S \cup \{x\}$, that is, $L_A = qH - U$, and $L_A + qj = q(j+H) - U$. Conjugating by the diagonal matrix $(-1)^{|S|}$ turns $-U$ into U. Hence the summand contributes $\operatorname{coker}(q(j+H) + U \text{ on } B_{d-j}) = X_q(d-j, j)$, $\binom{d}{j}(q-2)^j$ times. □

Proof of Corollary 1. By Theorem A it suffices to see that $\operatorname{Syl}_p X_q(m,c)$ depends only on the p-adic valuations of the diagonal entries $a_s = q(c+s)$. This is Lemma 2. □

Lemma 2 (only the valuations matter)**.** *Let $a_0, \dots, a_m$ be elements of $\mathbb{Z}_{(p)}$ and $M = \operatorname{diag}(a_{|S|}) + U$ on $B_m \otimes \mathbb{Z}_{(p)}$. If $a_s = p^{w_s} u_s$ with units u_s (or $a_s = 0$, with $u_s = 1$), then M is equivalent over $\mathbb{Z}_{(p)}$ to*

$$\operatorname{diag}(p^{w_{|S|}}) + U \qquad (\textit{with } p^w := 0 \textit{ when } a_s = 0).$$

Proof. Look for units λ_s, μ_s such that $P = \operatorname{diag}(\lambda_{|S|})$ and $Q = \operatorname{diag}(\mu_{|S|})$ give $PMQ = \operatorname{diag}(p^{w_{|S|}}) + U$. The diagonal of PMQ on rank s is $\lambda_s a_s \mu_s$ and its block from rank s to rank $s+1$ is $\lambda_{s+1}\mu_s U$, so we need $\lambda_s \mu_s u_s = 1$ and $\lambda_{s+1}\mu_s = 1$. Put $\mu_0 = 1$, $\lambda_0 = u_0^{-1}$, and recursively $\lambda_{s+1} = \mu_s^{-1}$, $\mu_{s+1} = (\lambda_{s+1} u_{s+1})^{-1}$. □

3. Strings, and the primes $p > m$

On B_m one has $[D, U] = m - 2H$: on a set S of size s, $DU(S) = (m-s)S + \Sigma$ and $UD(S) = s\,S + \Sigma$ with the same sum $\Sigma = \sum_{y \in S,\, x \notin S} (S \setminus \{y\}) \cup \{x\}$. For $0 \leq k \leq m/2$ let $\Lambda_k = \ker D \cap B_m(k)$. By induction on i, for $v \in \Lambda_k$,

$$D\,U^i v = i\,(m - 2k - i + 1)\,U^{i-1} v. \tag{1}$$

Over $\mathbb{Q}$ this is the classical $\mathfrak{sl}_2$-structure of the Boolean lattice [Pro82]:

$$B_m \otimes \mathbb{Q} = \bigoplus_{k \leq m/2} \bigoplus_{i=0}^{m-2k} U^i \Lambda_k^{\mathbb{Q}},$$

$\dim \Lambda_k^{\mathbb{Q}} = \binom{m}{k} - \binom{m}{k-1}$, and $U^{m-2k+1}\Lambda_k = 0$. We call $\Lambda_k, U\Lambda_k, \dots, U^{m-2k}\Lambda_k$ a *string*; its ranks are $k, \dots, m-k$ and its length is $m - 2k + 1$.

Lemma 3 (strings over $\mathbb{Z}_{(p)}$)**.** *If $p > m$, then $B_m \otimes \mathbb{Z}_{(p)} = \bigoplus_{k \leq m/2} \bigoplus_{i=0}^{m-2k} U^i \Lambda_k$, each Λ_k is free of rank $\binom{m}{k} - \binom{m}{k-1}$, and $U^i \colon \Lambda_k \to U^i \Lambda_k$ is an isomorphism for $i \leq m - 2k$.*

Proof. All lattices below are over $R = \mathbb{Z}_{(p)}$. First, for $r \le m/2$ we show $B_m(r) = UB_m(r-1) \oplus \Lambda_r$, by induction on r (for $r = 0$, $\Lambda_0 = B_m(0)$). By induction $B_m(r-1) = \bigoplus_{k<r} U^{r-1-k}\Lambda_k$, and by (1) the operator DU acts on $U^{r-1-k}\Lambda_k$ as the scalar $(r-k)(m-k-r+1)$; both factors lie in $[1, m]$, so they are units. Hence DU is invertible on $B_m(r-1)$, $\pi = U(DU)^{-1}D$ is an idempotent on $B_m(r)$ with image $UB_m(r-1)$, and its kernel is $\ker D \cap B_m(r) = \Lambda_r$ because $U(DU)^{-1}$ is injective. This gives the decomposition, and Λ_r is a direct summand, so free, of rank $\binom{m}{r} - \binom{m}{r-1}$.

Now fix any rank r and consider $\Phi\colon \bigoplus_{k\le \min(r,m-r)} \Lambda_k \to B_m(r)$, $(v_k) \mapsto \sum_k U^{r-k}v_k$. For $k' \le r$ let $\psi_{k'}\colon B_m(r) \to \Lambda_{k'}$ be $D^{r-k'}$ followed by the projection of $B_m(k') = UB_m(k'-1) \oplus \Lambda_{k'}$ onto $\Lambda_{k'}$. By (1), $D^{r-k'}U^{r-k}v_k$ is 0 if $k > k'$ (since $Dv_k = 0$), lies in $U^{k'-k}\Lambda_k \subseteq UB_m(k'-1)$ if $k < k'$, and equals $\prod_{i=1}^{r-k} i(m-2k-i+1)\, v_k$ if $k = k'$, a unit multiple of v_k (all factors lie in $[1, m]$ because $k \le m-r$). So $(\psi_{k'}) \circ \Phi$ is triangular with unit diagonal, Φ is split injective, and its image is a direct summand of $B_m(r)$ of full rank (compare ranks over $\mathbb{Q}$). Hence Φ is an isomorphism. Finally $U^{m-2k+1}\Lambda_k = 0$ holds over $\mathbb{Q}$, hence over R. □

Each string is stable under $q(c+H)+U$: in the basis $v, Uv, \dots, U^{\ell-1}v$ ($\ell = m-2k+1$) the operator is lower bidiagonal with diagonal entries $q(c+k+i)$, $0 \le i < \ell$, and all subdiagonal entries 1. The cokernel of such a matrix over a principal ideal domain is cyclic of order its determinant, because the minor obtained by deleting the first row and last column equals 1; if one diagonal entry is 0 (only possible for $c = k = i = 0$) the cokernel is $\mathbb{Z}_{(p)}$. This proves:

Theorem E (pieces with $m < p$)**.** *Let p be any prime, $e = v_p(q)$, and $m < p$. Then*

$$\operatorname{Syl}_p X_q(m,c) \;\cong \bigoplus_{\substack{0\le k\le m/2\\ (c,k)\neq(0,0)}} \big(\mathbb{Z}/p^{A_k}\big)^{\binom{m}{k}-\binom{m}{k-1}}, \qquad A_k = \sum_{r=k}^{m-k} \big(e + v_p(c+r)\big),$$

and for $c = 0$ the string $k = 0$ contributes only the free summand $\mathbb{Z}$.

Proof of Theorem B. If $p > d$, every piece $X_q(d-j, j)$ has $m = d-j < p$ and $c+r \le d < p$, so $v_p(c+r) = 0$ for $c+r \ge 1$ and $A_k = e(m-2k+1)$. Put $v = m-2k+1 = d+1-j-2k$; then $\binom{m}{k} - \binom{m}{k-1} = \binom{v+2k-1}{k} - \binom{v+2k-1}{k-1} = \frac{v}{v+2k}\binom{v+2k}{k} = b(k,v)$. Summing over j with Theorem A gives the formula. The generating function follows from $\sum_k b(k,v)x^k = C(x)^v$. □

4. The exponent

Lemma 4. *Let $c \ge 1$ and $M = q(c+H)+U$ on B_m. Then M is invertible over $\mathbb{Q}$, and for $S \subseteq T$ with $|S| = s$, $|T| = s+k$,*

$$(M^{-1})_{T,S} = (-1)^k\, k! \prod_{i=0}^{k} \frac{1}{q(c+s+i)},$$

while $(M^{-1})_{T,S} = 0$ if $S \not\subseteq T$.

Proof. With $\Delta = q(c+H)$, $M^{-1} = (1+\Delta^{-1}U)^{-1}\Delta^{-1} = \sum_{k\ge 0}(-\Delta^{-1}U)^k\Delta^{-1}$, a finite sum. A path from S to T in $(\Delta^{-1}U)^k\Delta^{-1}$ is a maximal chain $S = S_0 \subset \cdots \subset S_k = T$, there are $k!$ of them, and each collects Δ^{-1} once at every rank from s to $s+k$. □

For a nonsingular square matrix over a discrete valuation ring with uniformizer p, the exponent of the cokernel is p^ϵ with $-\epsilon$ the least valuation of an entry of M^{-1} (write M in Smith form). Hence, for $c \ge 1$, the exponent of $\operatorname{Syl}_p X_q(m,c)$ is p to the power

$$\max_{\substack{s,k\ge 0\\ s+k\le m}} \Big(\sum_{i=0}^{k} \big(e + v_p(c+s+i)\big) - v_p(k!)\Big) = \max_{c\le a\le b\le c+m} \Big(eL + v_p(L) + v_p\tbinom{b}{L}\Big), \quad L = b-a+1,$$

where $a = c+s$, $b = c+s+k$, and we used $\frac{b!}{(a-1)!\,(b-a)!} = L\binom{b}{L}$.

Lemma 5 (carries)**.** *For $d \geq 1$ and $e \geq 1$,*

$$\max_{1 \leq L \leq b \leq d} \Big(eL + v_p(L) + v_p\tbinom{b}{L} \Big) = \max_{1 \leq s \leq d} \big(es + v_p(s) \big),$$

and for $e = 0$ the left side equals $\lfloor \log_p d \rfloor$.

Proof. Taking $L = b = s$ gives $\geq$. Let $\lambda = \lfloor \log_p b \rfloor$. By Kummer's theorem $v_p\binom{b}{L}$ is the number of carries in the addition $L + (b - L)$ in base p. No carry occurs at a position below $v_p(L)$, where the digits of L vanish, and a carry out of position i forces $b \geq p^{i+1}$, so $i \leq \lambda - 1$. Hence $v_p(L) + v_p\binom{b}{L} \leq \lambda$. For $e = 0$ this gives the bound $\lfloor \log_p d \rfloor$, attained at $L = b = p^{\lfloor \log_p d \rfloor}$.

Let $e \geq 1$; we induct on b, the case $b = 1$ being clear. Let $s^* = p^\lambda \lfloor b/p^\lambda \rfloor$, so $1 \leq s^* \leq b < s^* + p^\lambda$ and $v_p(s^*) \geq \lambda$. If $L \leq s^*$, then $eL + v_p(L) + v_p\binom{b}{L} \leq es^* + \lambda \leq es^* + v_p(s^*)$. If $L > s^*$, write $b = s^* + r$ and $L = s^* + r'$ with $1 \leq r' \leq r < p^\lambda$. Then $v_p(L) = v_p(r')$, and since $(b - L) + r' = r < p^\lambda$, the additions $L + (b - L)$ and $r' + (r - r')$ have the same carries, so $v_p\binom{b}{L} = v_p\binom{r}{r'}$. By induction there is $1 \leq t \leq r$ with $er' + v_p(r') + v_p\binom{r}{r'} \leq et + v_p(t)$. Since $t < p^\lambda \leq p^{v_p(s^*)}$, $v_p(s^* + t) = v_p(t)$, and so $eL + v_p(L) + v_p\binom{b}{L} \leq e(s^* + t) + v_p(s^* + t)$ with $s^* + t \leq b$. □

Proof of Theorem C. By Theorem A the exponent is the largest exponent of the pieces $\mathrm{Syl}_p X_q(d - j, j)$ that occur. For $j \geq 1$ the piece occurs ($q \geq 3$) and the computation above gives the maximum over $j \leq a \leq b \leq d$; for $j = 1$ this is the full range $1 \leq a \leq b \leq d$. For $j = 0$, the row of the empty set in $M_0 = qH + U$ is zero and M_0 maps into $B' = \bigoplus_{s \geq 1} B_d(s)$, so

$$\operatorname{coker} M_0 = \mathbb{Z} \oplus B'/(M''B' + \mathbb{Z}u),$$

where M'' is the restriction of M_0 to B' (nonsingular, lower triangular by rank with diagonal qs) and $u = M_0(\emptyset)$. Its torsion is a quotient of $\operatorname{coker} M''$, whose exponent is given by Lemma 4 (with $c = 0$ and $s \geq 1$) as the same maximum over $1 \leq a \leq b \leq d$. So the exponent is p to $\max_{1 \leq a \leq b \leq d}(eL + v_p(L) + v_p\binom{b}{L})$, and Lemma 5 finishes. □

For $q = 2$ the same argument gives only an upper bound, and it is not attained: for Q_2 the group is $\mathbb{Z}/4$ while the bound is 2^3.

5. The number of cyclic factors

Proof of Theorem D. The number of cyclic factors of $\mathrm{Syl}_p K$ is $\dim_{\mathbb{F}_p}(\operatorname{coker} L \otimes \mathbb{F}_p) - 1$, the 1 for the free summand. Modulo p, $q(j + H) + U \equiv U$, so by Theorem A $\dim_{\mathbb{F}_p}(\operatorname{coker} L \otimes \mathbb{F}_p) = \sum_j \binom{d}{j}(q - 2)^j \big(2^{d-j} - \operatorname{rank}_{\mathbb{F}_p} U|_{B_{d-j}}\big)$. The rank of U is the sum over s of the ranks of the inclusion matrices $W_{s,s+1}$ of s-subsets versus $(s + 1)$-subsets of $[m]$. By Wilson's diagonal form [Wil90], for $t \leq \min(k, m - k)$ the matrix $W_{t,k}$ is equivalent over $\mathbb{Z}$ to a diagonal matrix with entries $\binom{k-i}{t-i}$ of multiplicity $\binom{m}{i} - \binom{m}{i-1}$, $0 \leq i \leq t$ (we use the statement as given in [CSX15, (2)]); so its rank over $\mathbb{F}_p$ is $\sum_{i:\, p \nmid \binom{k-i}{t-i}} \big(\binom{m}{i} - \binom{m}{i-1}\big)$. For $W_{s,s+1}$ with $2s \leq m - 1$ take $(t, k) = (s, s + 1)$, so $\binom{k-i}{t-i} = s + 1 - i$; otherwise complementation identifies $W_{s,s+1}$ with the transpose of $W_{m-s-1,m-s}$ and we take $(t, k) = (m - 1 - s, m - s)$. This is $r_p(m, s)$. □

6. The range $p \leq d$: the Glue Conjecture

By Theorem E the only pieces not yet described are $X_q(m, c)$ with $m \geq p$. For $p \leq m$ the strings of length $> p$ break: by (1) the coefficient $i(m - 2k - i + 1)$ vanishes modulo p. For $m \leq 2p - 2$ every broken string meets exactly one other string. In the language of the representations of SL_2 in characteristic p: $B_m \otimes \mathbb{F}_p = (\mathbb{F}_p^2)^{\otimes m}$ is a tilting module, and for $p \leq \lambda \leq 2p - 2$ the indecomposable tilting module $T(\lambda)$ has Weyl factors of highest weights λ and $2p - 2 - \lambda$ [Don93, Jan03]. The conjecture says that the glue between these two strings moves exactly one factor p from the long string to the short one.

For $\lambda \equiv m \pmod 2$, $0 \le \lambda \le m$, write $\beta(m,\lambda) = \binom{m}{(m-\lambda)/2} - \binom{m}{(m-\lambda)/2-1}$ (the number of strings of length $\lambda+1$), and

$$A_c(m,\lambda) = \sum_{r=(m-\lambda)/2}^{(m+\lambda)/2} \big(e + v_p(c+r)\big),$$

the valuation of the determinant of $q(c+H)+U$ on one such string.

Conjecture F (the Glue Conjecture)**.** *Let p be a prime, q an integer with $e = v_p(q) \ge 1$, $m \le 2p-2$ and $c \ge 0$. Then*

$$\begin{aligned}\operatorname{Syl}_p X_q(m,c) \;\cong\; &\bigoplus_{\lambda \le p-1} \big(\mathbb{Z}/p^{A_c(m,\lambda)}\big)^{\beta(m,\lambda)-\beta'(m,\lambda)} \\ &\oplus \bigoplus_{p \le \lambda \le m} \Big(\mathbb{Z}/p^{A_c(m,\lambda)-1} \oplus \mathbb{Z}/p^{A_c(m,2p-2-\lambda)+1}\Big)^{\beta(m,\lambda)},\end{aligned}$$

where $\beta'(m,\lambda) = \beta(m,2p-2-\lambda)$ if $\lambda \le p-2$ and $\beta'(m,p-1) = 0$, and λ runs over integers $\equiv m \pmod 2$. When $c = 0$ the string of length $m+1$ (through the empty set) has a zero on its diagonal: for $\lambda = m < p$ it gives nothing, and for $\lambda = m \ge p$ the term $\mathbb{Z}/p^{A_c(m,m)-1}$ is omitted (it is the free summand) while its partner $\mathbb{Z}/p^{A_0(m,2p-2-m)+1}$ stays.

For $m < p$ the conjecture is Theorem E. Together with Theorem A it determines $\operatorname{Syl}_p K(H(d,q))$ for every $d \le 2p-2$. The hypothesis $e \ge 1$ is needed: for $p \nmid q$ the formula fails, for instance when $(q,p,d) = (4,3,4)$; there the answer of [JNR03] applies instead.

Example 6. For $p = q = 3$ and $d = 4$: Theorem A gives the pieces $X(4,0)$, $X(3,1)^4$, $X(2,2)^6$, $X(1,3)^4$, $X(0,4)$, and the conjecture gives $\operatorname{Syl}_3 K(H(4,3)) \cong (\mathbb{Z}/3)^2 \oplus (\mathbb{Z}/3^2)^7 \oplus (\mathbb{Z}/3^3)^8 \oplus (\mathbb{Z}/3^4)^{17}$, which is the group we computed.

7. Verification

All computations were exact. The Smith forms over $\mathbb{Z}_{(p)}$ were computed modulo p^M with $p^M < 2^{31}$, by unit pivots and Schur complements, level by level in the p-adic valuation; factors of valuation at least M were then determined separately, by the matrix $p^E M^{-1}$ of Lemma 4, whose elementary divisors are p^{E-a} for the divisors p^a of M. The scripts, the sealed prediction files and the logs are at `https://github.com/tretoef-estrella/coloured-sand-theorem` and in the Zenodo record doi:10.5281/zenodo.23327609.

(1) Theorem A: the Sylow p-subgroups for $p \in \{2,3,5,7\}$ computed directly from the Laplacian and from the pieces agree in all 90 comparisons (27 cells, $q \in \{3,\dots,10,12\}$, $q^d \le 729$, including $p \mid q$ and $p \le d$); adding one extra copy of a piece changes the answer in 49 of them. Independently, the cells $(d,q) = (3,7)$ and $(2,11)$ were recomputed with Sage from the full Laplacian.

(2) Theorem B: found first as a conjecture from the cells with $d \le 4$; before the proof, ten predictions for $(d,p) = (2,11),(2,13),(2,17),(2,19),(2,23),(3,17),(3,19),(3,23),(3,29),(4,13)$ were written to a file (MD5 `d5c36be3736079cbc2349f469751c627`) and then computed: ten agreements.

(3) Theorem C: 143 cells, for $q \in \{3,\dots,10,12\}$ and $p \in \{2,3,5,7\}$, both with $p \mid q$ (including $p = 2$) and with $p \nmid q$, all in agreement; the guess $ed + v_p(d)$ fails in one of them and the guess ed in 26.

(4) Theorem D: the rank formula for $W_{s,s+1}$ checked directly in 264 cases ($p \le 7$, $m \le 11$); the value of N agrees with the computed groups in 23 cells with $p \le d$.

(5) Conjecture F: found from 88 pieces with $p = 3,5$. Then 30 pieces with $p = 7$, $7 \le m \le 11$, $1 \le c \le 6$ were predicted before being computed (MD5 of the file `1f5cc12e88f12e375022658da3eea6ce`): 30 agreements. Then 14 full groups $\operatorname{Syl}_p K(H(d,q))$ with composite $q \in \{6,9,10,12,15,20,25,27\}$ and $p \le d \le 2p-2$ (MD5 `341a2c48`

`9aef1c30349bae079ddcce81`): 14 agreements. It also agrees with the 10 full groups computed for $q = p \in \{3, 5, 7\}$ and $p \le d \le 2p - 2$.

8. Data beyond $2p - 2$

For $q = p = 3$ the computed groups, written as $(n_1, n_2, \dots)$ with n_v the number of factors $\mathbb{Z}/3^v$, are:

d	$(n_1, n_2, \dots)$
5	$(12, 10, 16, 53, 5)$
6	$(38, 37, 36, 117, 24, 16, 6)$
7	$(86, 105, 127, 285, 85, 63, 41)$
8	$(250, 211, 286, 1078, 174, 133, 174, 8)$
9	$(942, 412, 624, 3339, 449, 292, 495, 133, 84, 37, 9)$

For $p = 2$ the whole problem is the hypercube, solved in [DST] with the tilting modules of SL_2 in characteristic 2; we expect the general case $m \ge 2p - 1$ to follow the same pattern, with tilting modules having more Weyl factors.

Provenance

The results were found on the night of 12–13 October 2026 by an AI system, Claude (Anthropic), working for the author: it chose the problem after a search of the literature, wrote the computations, found the formulas, sealed and checked the predictions, and found the proofs; it also found, when checking the literature before publication, that Theorem A is [JNR03, Theorem 13]. The author takes responsibility for the paper. To the best of our knowledge, after a search of the works citing [JNR03] and of arXiv, Theorems B, C, D and E and Conjecture F are new; Theorem A is due to Jacobson, Niedermaier and Reiner.

References

[Bai03] H. Bai, On the critical group of the n-cube, *Linear Algebra Appl.* 369 (2003), 251–261.

[Big99] N. L. Biggs, Chip-firing and the critical group of a graph, *J. Algebraic Combin.* 9 (1999), 25–45.

[CSX15] D. B. Chandler, P. Sin, Q. Xiang, The Smith group of the hypercube graph, arXiv:1511.00272.

[DJ14] J. E. Ducey, D. M. Jalil, Integer invariants of abelian Cayley graphs, *Linear Algebra Appl.* 445 (2014), 316–325; arXiv:1308.2335.

[Don93] S. Donkin, On tilting modules for algebraic groups, *Math. Z.* 212 (1993), 39–60.

[DST] R. Amichis Luengo, The Dancing Sand Theorem: the 2-part of the sandpile group of the hypercube, for every n, preprint (2026), hexagon:2610.00205; doi:10.5281/zenodo.23188298.

[Jan03] J. C. Jantzen, *Representations of Algebraic Groups*, 2nd ed., Math. Surveys Monogr. 107, Amer. Math. Soc., 2003.

[JNR03] B. Jacobson, A. Niedermaier, V. Reiner, Critical groups for complete multipartite graphs and Cartesian products of complete graphs, *J. Graph Theory* 44 (2003), 231–250.

[Pro82] R. A. Proctor, Representations of $\mathfrak{sl}(2, \mathbb{C})$ on posets and the Sperner property, *SIAM J. Algebraic Discrete Methods* 3 (1982), 275–280.

[Wil90] R. M. Wilson, A diagonal form for the incidence matrices of t-subsets vs. k-subsets, *European J. Combin.* 11 (1990), 609–615.

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